Akshay Popat

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INTERESTS

Algebraic topology, homotopy theory, algebraic geometry and representation theory

EDUCATION

Imperial College London, Mathematics 10/2023 – present

- 3rd year average: 80.7
- Currently taking: Manifolds, Algebra 4, Commutative algebra
- Select 3rd year modules: Algebraic topology (91), Algebraic number theory (91), Galois theory (87), Group representation theory (86)

Tiffin School 09/2015 – 06/2022

- A-Levels: A*A*A* (Mathematics, Further mathematics, Physics)

PROJECTS

Hitchin systems and their quantization 10/2026 – present

- 4th year project supervised by Professor Yankı Lekili

EPSRC Vacation Internship – Principal bundles 09/2026

- Selected for an EPSRC-funded internship supervised by Dr Marie-Amelie Lawn
- Proved the equivalence between rank n real vector bundles and principal $GL_n(\mathbb{R})$-bundles, and studied their classification via the infinite Grassmannian $Gr_n(\mathbb{R}^\infty)$
- Explicitly constructed examples of spin structures
- Proved that an $SO(n)$-bundle over a manifold M has a spin structure if and only if the second Stiefel-Whitney class vanishes, in which case the spin structures are classified by $H^1(M;\mathbb{Z}_2)$

Multilinear Kakeya Estimates 06/2025

- 2nd year group project supervised by Dr Zhiyuan Zhang
- Proved the Borsuk-Ulam theorem and learned of its application in proving a weak Bennett-Carbery-Tao multilinear Kakeya estimate

TECHNICAL SKILLS

LaTeX, Python

TUTORING

We Solve Problems – KCL Maths Circle Tutor 01/2026 – present

- Help students through material and problems

ADDITIONAL

Imperial College MathsWiki 10/2024 – present

- Contributing to making notes for the undergraduate courses at Imperial

Japanese 07/2022 – present

- Self-studying Japanese
- N1 level